Tharun Ganeshram

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Technical Skills

Languages & Frameworks: C, C++, Python, Bash, Expect-Lite (TCL extension), MLIR, Intel TBB, CUDA
OS & Tools: Linux, FreeRTOS, Sysfs, IAR, MATLAB, Git, Jira, Confluence, SolidWorks, PlantUML, InfluxDB
Electrical & Hardware: SPI, CAN, UART, I2C, STM32, ESP32, Arduino, Raspberry Pi, Pixhawk, Circuit Design

Education

University of Waterloo — **Sep. 2022 – Apr. 2027**
*Candidate for Bachelor of Applied Science (B.A.Sc.) in **Honours Systems Design Engineering*** — *GPA: 4.0*

Experience

NVIDIA — **Sep. 2025 – Dec. 2025**
Embedded Systems Software Engineering Intern — *Santa Clara, CA*
- Incoming Fall 2025

Cerebras Systems — **Jan. 2025 – Apr. 2025**
Machine Learning Software Engineering Intern — *Toronto, ON*
- Redesigned inference pipeline idle logic using placeholder tokens to **reduce idle sleep & startup delays by 100%**
- **Exponentially increased LLM speculative decoding metrics** via **C++** per-token state tracking & **InfluxDB** output
- Designed prefix search in **Python** for off-wafer prompt caching that **increased inference speed by over 40%**
- Pioneered **multimodal LLM inference** runtime compatibility while maintaining **existing time** & **space** complexity
- Enabled **16+** on-wafer boolean operations by implementing custom **MLIR** compiler passes, operations, & lowerings

University of Waterloo Hyperloop Design Team — **Dec. 2022 – Apr. 2025**
Firmware Team Lead — *Waterloo, ON*
- **Achieved sub-100ms latency** after system re-architecture using **FreeRTOS** thread & resource management
- Fulfilled battery safety compliance by configuring thermistors and fans to **STM32** & **Raspberry Pi** using **SPI** & **CAN**

Waterloo Aerial Robotics Group — **May 2024 – Dec. 2024**
Embedded Flight Software Developer — *Waterloo, ON*
- Architected **37% signal noise reduction** by investigating **STM32 DroneCAN** integration, decoding, & debugging
- Established the conversion of **DroneCAN** servo signals from a **Pixhawk** to PWM motor signals using **STM32** in **C**

Blackline Safety — **May 2024 – Aug. 2024**
Firmware Engineering Intern — *Waterloo, ON*
- Implemented **UART** communication, **BSP** control signal abstraction & sensor classes in **C++** for **5** new EXO8 sensors
- Configured execute-in-place flash memory & clock with screen optimizations for **6x increase** in XIP processing rate

Ciena — **Jan. 2023 – Apr. 2023**
Embedded Software Engineering Intern — *Ottawa, ON*
- Engineered **Expect-Lite** to **Python** converter, reducing manual file upgrade time from **3 weeks to 5 hours**
- Modernized legacy **Bash** Sanity systems using **Python**, redesigning logic with modules to **reduce runtime by 80%**

Projects

Autonomous System Monitor & Dynamic Resource Manager | *Embedded C, Linux Kernel, Sysfs, Character Device*
- Engineered a **Linux kernel module** with **HRTimer**-driven workload monitoring, dynamic resource factor adjustment, and user-kernel interaction via **character device** & **Sysfs**

Real-Time Object Recognition App | *Python, TensorFlow, OpenCV*
- Trained unsupervised ML model using **TensorFlow** & **OpenCV** to accurately identify & classify objects in live video

Omni-Directional Bluetooth Car | *Embedded C, STM32, ESP32, UART, SolidWorks*
- Designed car body & firmware using **servo motors**, **STM32**, and **ESP32** for real-time **UART Bluetooth** control